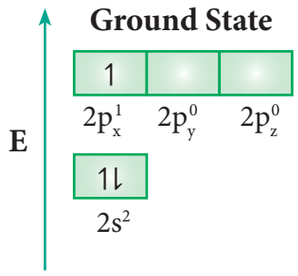
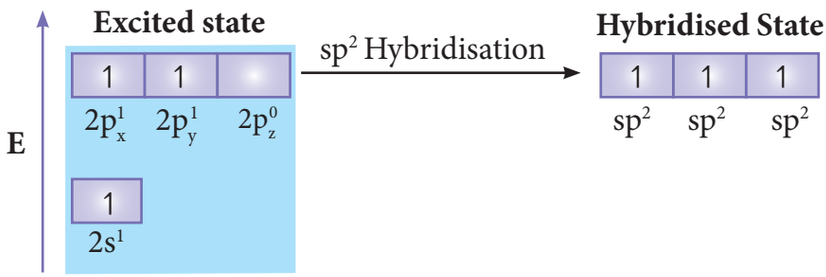
sp^2 Hybridisation:

Consider the bond formation in boron trifluoride.The ground state valence shell electronic configuration of Boron atom is $[He]2s^2\ 2p^1$.



In the ground state boron has only one unpaired electron in the valence shell. In order to form three covalent bonds with fluorine atoms, three unpaired electrons are required. To achieve this, one of the paired electrons in the 2s orbital is promoted to the $2p_y$ orbital in the excite state.

In boron, the s orbital and two p orbitals (p_x and p_y) in the valence shell hybridses, to generate three equivalent sp^2 orbitals as shown in the Figure. These three orbitals lie in the same xy plane and the angle between any two orbitals is equal to 120^0



Overlap with 2pz orbitals of fluorine:

The three sp^2 hybridised orbitals of boron now overlap with the $2p_z$ orbitals of fluorine (3 atoms). This overlap takes place along the axis as shown below.

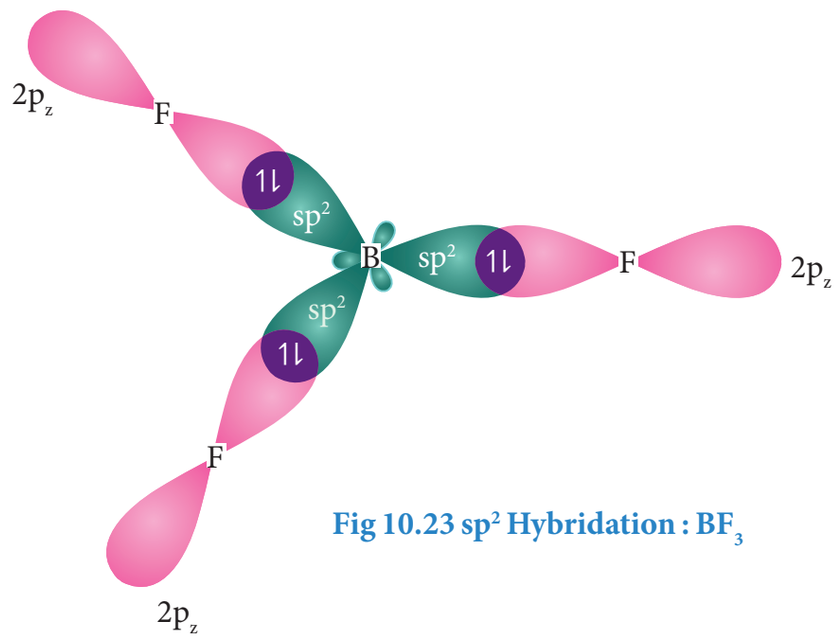


Fig 10.23 sp^2 Hybridation : BF_3

11th Chemistry Chapter-10 Vol-2 EM.indd 91 09-01-2023 14:39:44